

11th grade

Learning evidence 2.1: Confidence Interval Planning (Solutions)

Problems

Problem 1: Problem description

A city budgeting office wants to estimate the average weekly transit subsidy per rider for two pilot zones. Zone A recorded the following sample of weekly subsidy amounts (USD):

- Zone A sample ($n_A = 6$): 18, 22, 19, 21, 20, 17.
- Zone B sample ($n_B = 8$): 24, 26, 23, 25, 27, 22, 24, 26.

An annual audit provides the population standard deviations for the zones: $\sigma_A = 3,5$ and $\sigma_B = 4,0$. For classroom purposes, suppose the true population mean weekly subsidy is $\mu_A = 20$ dollars in Zone A and $\mu_B = 25$ dollars in Zone B. Construct and interpret a 95 % confidence interval for the population mean weekly subsidy in each zone.

C1

First, we calculate the sample means.

$$\bar{x}_A = \frac{\sum x_A}{n_A}$$

$$\sum x_A = 18 + 22 + 19 + 21 + 20 + 17 = 117$$

$$\bar{x}_A = \frac{117}{6} = 19,5$$

$$\bar{x}_B = \frac{\sum x_B}{n_B}$$

$$\sum x_B = 24 + 26 + 23 + 25 + 27 + 22 + 24 + 26 = 197$$

$$\bar{x}_B = \frac{197}{8} = 24,625$$

Second, we calculate the sample standard deviations.

$$s_A^2 = \frac{\sum (x_A - \bar{x}_A)^2}{n_A - 1}$$

$$\sum (x_A - \bar{x}_A)^2 = (18 - 19,5)^2 + (22 - 19,5)^2 + (19 - 19,5)^2 + (21 - 19,5)^2 + (20 - 19,5)^2 + (17 - 19,5)^2 = 17,5$$

$$s_A = \sqrt{\frac{17,5}{6 - 1}} = \sqrt{3,5} \approx 1,87$$

$$s_B^2 = \frac{\sum (x_B - \bar{x}_B)^2}{n_B - 1}$$

$$\sum (x_B - \bar{x}_B)^2 = (24 - 24,625)^2 + (26 - 24,625)^2 + (23 - 24,625)^2 + (25 - 24,625)^2 + (27 - 24,625)^2 + (22 - 24,625)^2 + (24 - 24,625)^2 + (26 - 24,625)^2 = 19,875$$

$$s_B = \sqrt{\frac{19,875}{8 - 1}} \approx 1,69$$

C2

The sample statistics are $\bar{x}_A = 19,5$, $s_A \approx 1,87$, $\bar{x}_B = 24,625$, and $s_B \approx 1,69$, while the population parameters are μ_A , μ_B , $\sigma_A = 3,5$, and $\sigma_B = 4,0$. The sample values estimate the population values.

C3

A single point estimate can miss the true mean because of sampling variation. A confidence interval shows a range of plausible values for each μ , which is more informative for planning.

C5

With known $\sigma_A = 3,5$ and $n_A = 6$,

$$SE_A = \frac{\sigma_A}{\sqrt{n_A}}$$
$$SE_A = \frac{3,5}{\sqrt{6}} \approx 1,43$$

For 95 % confidence, $z_{0,025} = 1,96$.

$$E_A = z_{0,025} \cdot SE_A$$
$$E_A = 1,96(1,43) \approx 2,80$$
$$\mu_A \in \bar{x}_A \pm E_A$$
$$\mu_A \in 19,5 \pm 2,80 = [19,5 - 2,80, 19,5 + 2,80] = [16,70, 22,30]$$

With known $\sigma_B = 4,0$ and $n_B = 8$,

$$SE_B = \frac{\sigma_B}{\sqrt{n_B}}$$
$$SE_B = \frac{4,0}{\sqrt{8}} \approx 1,41$$
$$E_B = z_{0,025} \cdot SE_B$$
$$E_B = 1,96(1,41) \approx 2,76$$
$$\mu_B \in \bar{x}_B \pm E_B$$
$$\mu_B \in 24,625 \pm 2,76 = [24,625 - 2,76, 24,625 + 2,76] = [21,87, 27,39]$$

C4

We are 95 % confident the true mean weekly subsidy in Zone A is between about 16.70 and 22.30 dollars, and 95 % confident the true mean weekly subsidy in Zone B is between about 21.87 and 27.39 dollars. The intervals reflect uncertainty from sampling in each zone.

Problem 2: Problem description

A regional warehouse studies the number of orders processed per hour to plan staffing. A random sample of 60 hours is summarized in grouped form below. The operations team also reports a known population standard deviation of $\sigma = 6,2$ orders per hour. For academic purposes only, assume the true population mean order rate is $\mu = 22$ orders per hour. The grouped data pairs (orders per hour, frequency) are: $14 \rightarrow 8$, $18 \rightarrow 16$, $22 \rightarrow 20$, $26 \rightarrow 10$, and $30 \rightarrow 6$. Construct and interpret a 95 % confidence interval for the population mean orders per hour.

C1

Using grouped values x with frequencies f .

$$\bar{x} = \frac{\sum fx}{n}$$

$$n = \sum f$$

$$n = 8 + 16 + 20 + 10 + 6 = 60$$

$$\sum fx = 8(14) + 16(18) + 20(22) + 10(26) + 6(30) = 1280$$

$$\bar{x} = \frac{1280}{60} \approx 21,33$$

Second, we calculate the sample standard deviation s .

$$s^2 = \frac{\sum f(x - \bar{x})^2}{n - 1}$$

$$\sum f(x - \bar{x})^2 \approx 8(14 - 21,33)^2 + 16(18 - 21,33)^2 + 20(22 - 21,33)^2 + 10(26 - 21,33)^2 + 6(30 - 21,33)^2$$

$$\sum f(x - \bar{x})^2 \approx 430,22 + 177,78 + 8,89 + 217,78 + 450,67 = 1285,34$$

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{n - 1}}$$

$$s = \sqrt{\frac{1285,34}{60 - 1}} \approx 4,67$$

C2

The sample statistics are $\bar{x} \approx 21,33$ and $s \approx 4,67$, while the population parameters are μ and $\sigma = 6,2$. The sample values estimate the population values.

C3

The sample summarizes only 60 hours, so the true mean could differ from the point estimate. A confidence interval expresses the likely range for μ based on the data.

C5

With known $\sigma = 6,2$ and $n = 60$,

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE = \frac{6,2}{\sqrt{60}} \approx 0,80$$

For 95 % confidence, $z_{0,025} = 1,96$.

$$E = z_{0,025} \cdot SE$$

$$E = 1,96(0,80) \approx 1,57$$

$$\mu \in \bar{x} \pm E$$

$$\mu \in 21,33 \pm 1,57 = [21,33 - 1,57, 21,33 + 1,57] = [19,76, 22,90]$$

C4

We are 95 % confident the true mean orders processed per hour is between about 19.76 and 22.90 orders. The interval shows the range supported by the sample and known standard deviation.

Problem 3: Problem description

A logistics firm monitors the time (in minutes) required to load containers at a port. A sample of 80 loading times is grouped into class intervals below. The firm has a historical estimate of the population standard deviation, $\sigma = 5,8$ minutes. For classroom reference, suppose the true population mean loading time is $\mu = 32$ minutes. The grouped intervals and frequencies are: 20–24 minutes (12), 25–29 minutes (18), 30–34 minutes (20), 35–39 minutes (17), and 40–44 minutes (13). Construct and interpret a 95 % confidence interval for the population mean loading time.

C1

Using grouped midpoints x with frequencies f .

$$\bar{x} = \frac{\sum fx}{n}$$

$$n = \sum f$$

$$n = 12 + 18 + 20 + 17 + 13 = 80$$

$$\sum fx = 12(22) + 18(27) + 20(32) + 17(37) + 13(42) = 2565$$

$$\bar{x} = \frac{2565}{80} = 32,0625$$

Second, we calculate the sample standard deviation s .

$$s^2 = \frac{\sum f(x - \bar{x})^2}{n - 1}$$

$$\sum f(x - \bar{x})^2 \approx 12(22 - 32,0625)^2 + 18(27 - 32,0625)^2 + 20(32 - 32,0625)^2 + 17(37 - 32,0625)^2 + 13(42 - 32,0625)^2$$

$$\sum f(x - \bar{x})^2 \approx 3374,69$$

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{n - 1}}$$

$$s = \sqrt{\frac{3374,69}{80 - 1}} \approx 6,54$$

C2

The sample statistics are $\bar{x} = 32,0625$ and $s \approx 6,54$, while the population parameters are μ and $\sigma = 5,8$. The sample values estimate the population values.

C3

Even with 80 observations, a point estimate does not show how much the mean could vary. A confidence interval summarizes that uncertainty by giving a plausible range for μ .

C5

With known $\sigma = 5,8$ and $n = 80$,

$$SE = \frac{\sigma}{\sqrt{n}}$$
$$SE = \frac{5,8}{\sqrt{80}} \approx 0,65$$

For 95 % confidence, $z_{0,025} = 1,96$.

$$E = z_{0,025} \cdot SE$$
$$E = 1,96(0,65) \approx 1,27$$

$$\mu \in \bar{x} \pm E$$

$$\mu \in 32,0625 \pm 1,27 = [32,0625 - 1,27, 32,0625 + 1,27] = [30,79, 33,33]$$

C4

We are 95 % confident the true mean loading time is between about 30.79 and 33.33 minutes. The interval provides a plausible range for the population mean based on the sample.